Computer Architecture & Organization

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Computer Architecture

- **Computer Architecture** is a functional description of requirements and design implementation for the various parts of a computer. It deals with the functional behavior of computer systems. It comes before the computer organization while designing a computer.

- Computer architecture and computer organization are related but distinct concepts in the field of computer science.

Computer Architecture

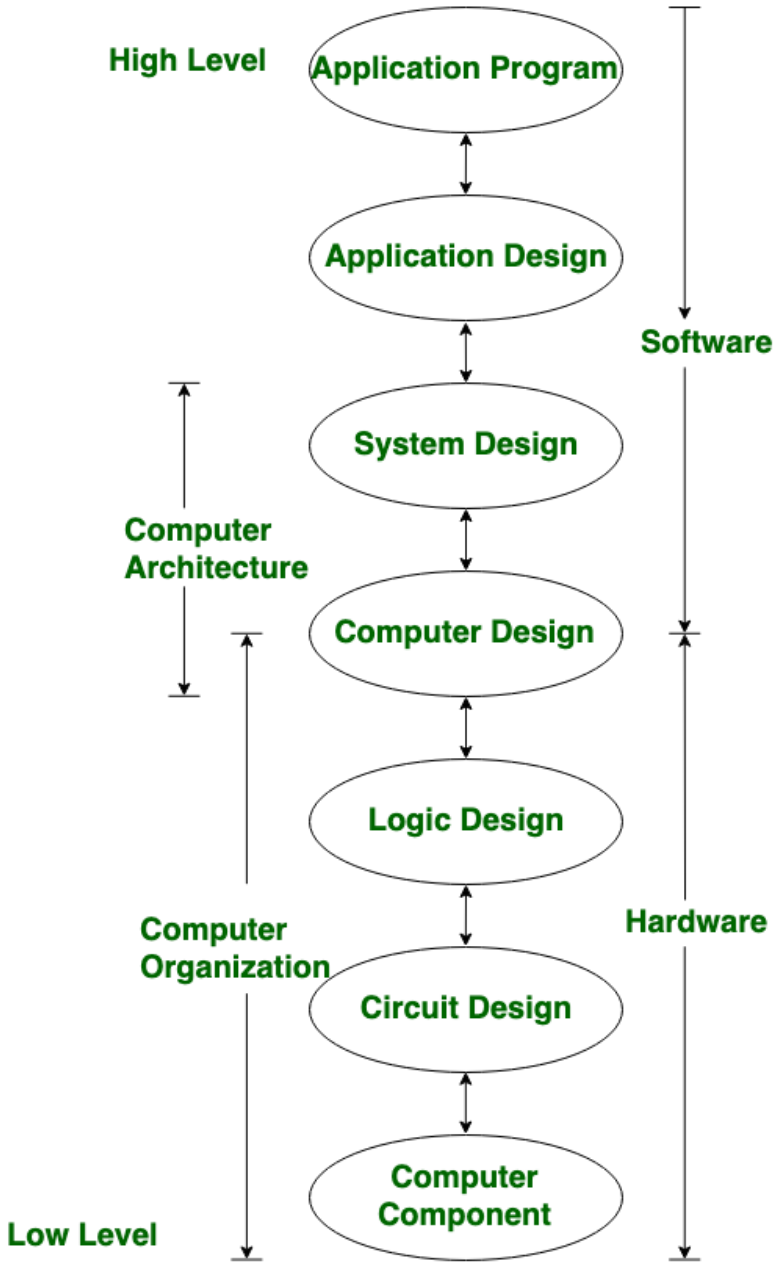
Computer Architecture:

1. Design of internal components like CPU and memory.
2. Involves decisions on instruction set, data path, and control unit.
3. Aims to optimize overall system performance.
4. Focuses on high-level design decisions.

Computer Architecture

Computer Organization:

1. Deals with operational units and their connections.
2. Concerned with physical implementation of architecture.
3. Involves decisions on interconnection, bus structure, and memory hierarchy.
4. Handles low-level implementation details.
5. Ensures components work together to perform operations efficiently.



Computer System

Difference between Computer Architecture and Computer Organization

Computer Architecture	Computer Organization
Architecture describes what the computer does.	The Organization describes how it does it.
Computer Architecture deals with the functional behavior of computer systems.	Computer Organization deals with a structural relationship.
It deals with high-level design issues.	It deals with low-level design issues.
Architecture indicates its hardware and software.	Whereas Organization indicates its performance.
As a programmer, you can view architecture as a series of instructions, addressing modes, and registers.	The implementation of the architecture is called organization.
For designing a computer, its architecture is fixed first.	For designing a computer, an organization is decided after its architecture.
Computer Architecture is also called Instruction Set Architecture (ISA).	Computer Organization is frequently called microarchitecture.

Difference between Computer Architecture and Computer Organization

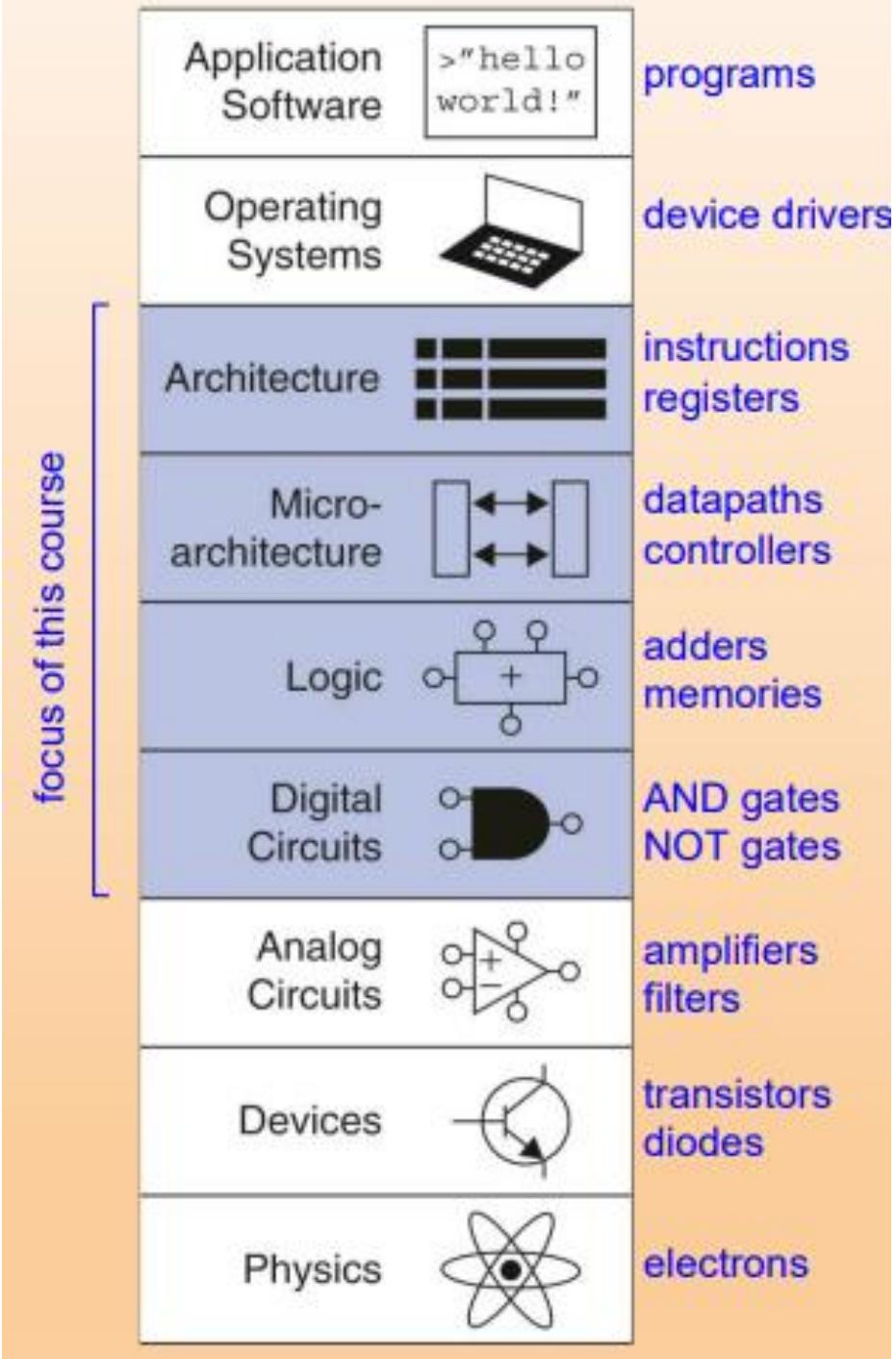
Computer Architecture	Computer Organization
Computer Architecture comprises logical functions such as instruction sets, registers, data types, and addressing modes.	Computer Organization consists of physical units like circuit designs, peripherals, and adders.
It makes the computer's hardware visible.	It offers details on how well the computer performs.
Architecture coordinates the hardware and software of the system.	Computer Organization handles the segments of the network in a system.
The software developer is aware of it.	It escapes the software programmer's detection.
Examples- Intel and AMD created the x86 processor. Sun Microsystems and others created the SPARC processor. Apple, IBM, and Motorola created the PowerPC.	Organizational qualities include hardware elements that are invisible to the programmer, such as interfacing of computer and peripherals, memory technologies, and control signals.

Architectural Attributes

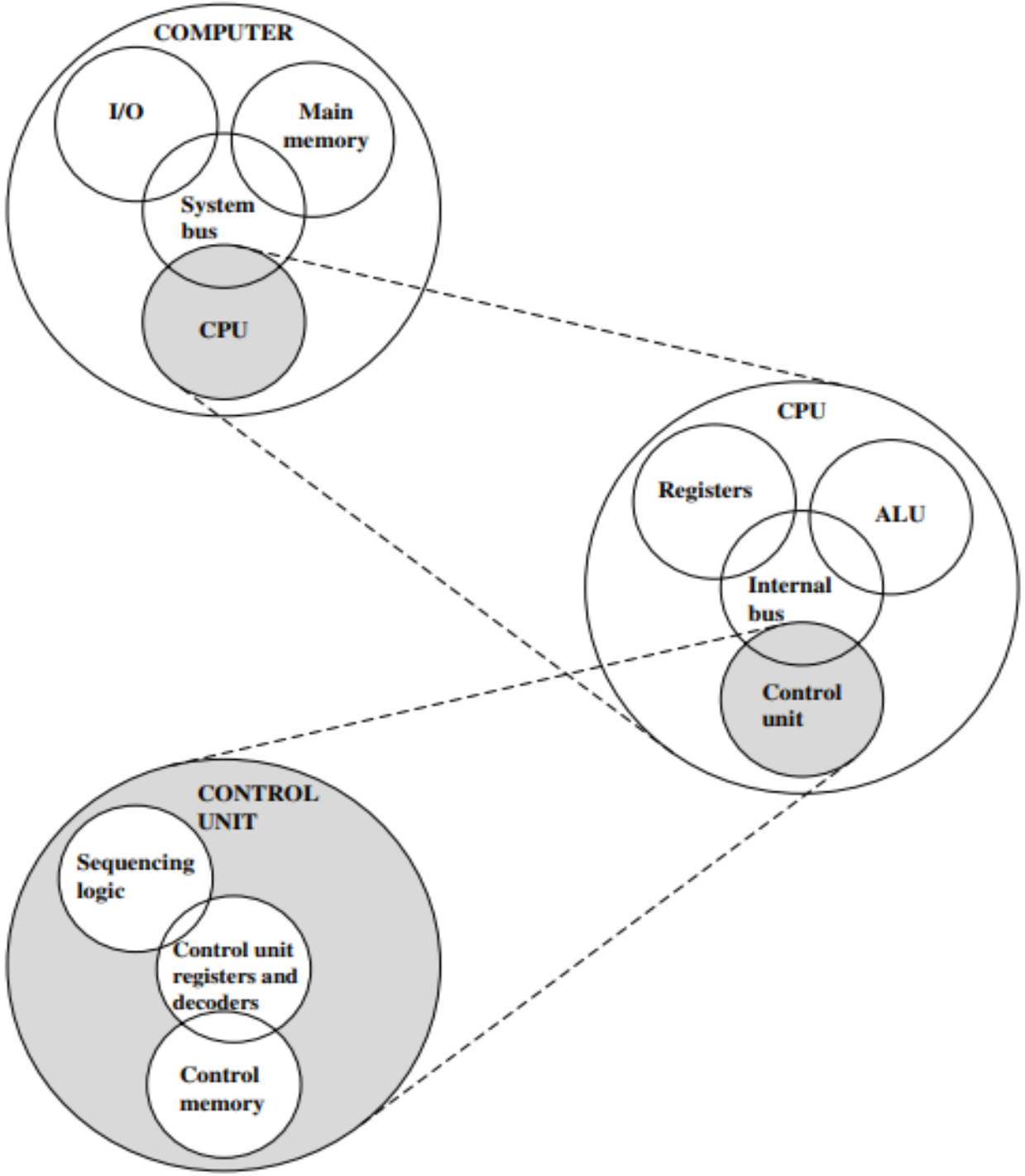
- I/O Mechanism
- Instruction Set (IS)
- Number of bits to represent various data types
- Addressing Memory

Organizational attributes

Hardware details transparent to the programmer, such as control signals; interfaces between the computer and peripherals; and the memory technology used.


Application Software
>"hello world!"
programs
Operating Systems
device drivers
Architecture
instructions
registers
Micro-architecture
datapaths
controllers
Logic
adders
memories
Digital Circuits
AND gates
NOT gates
Analog Circuits
amplifiers
filters
Devices
transistors
diodes
Physics
electrons
focus of this course

Top-Level Structure of a Computer



Top level View of Computer

1. Structural View
2. Functional View

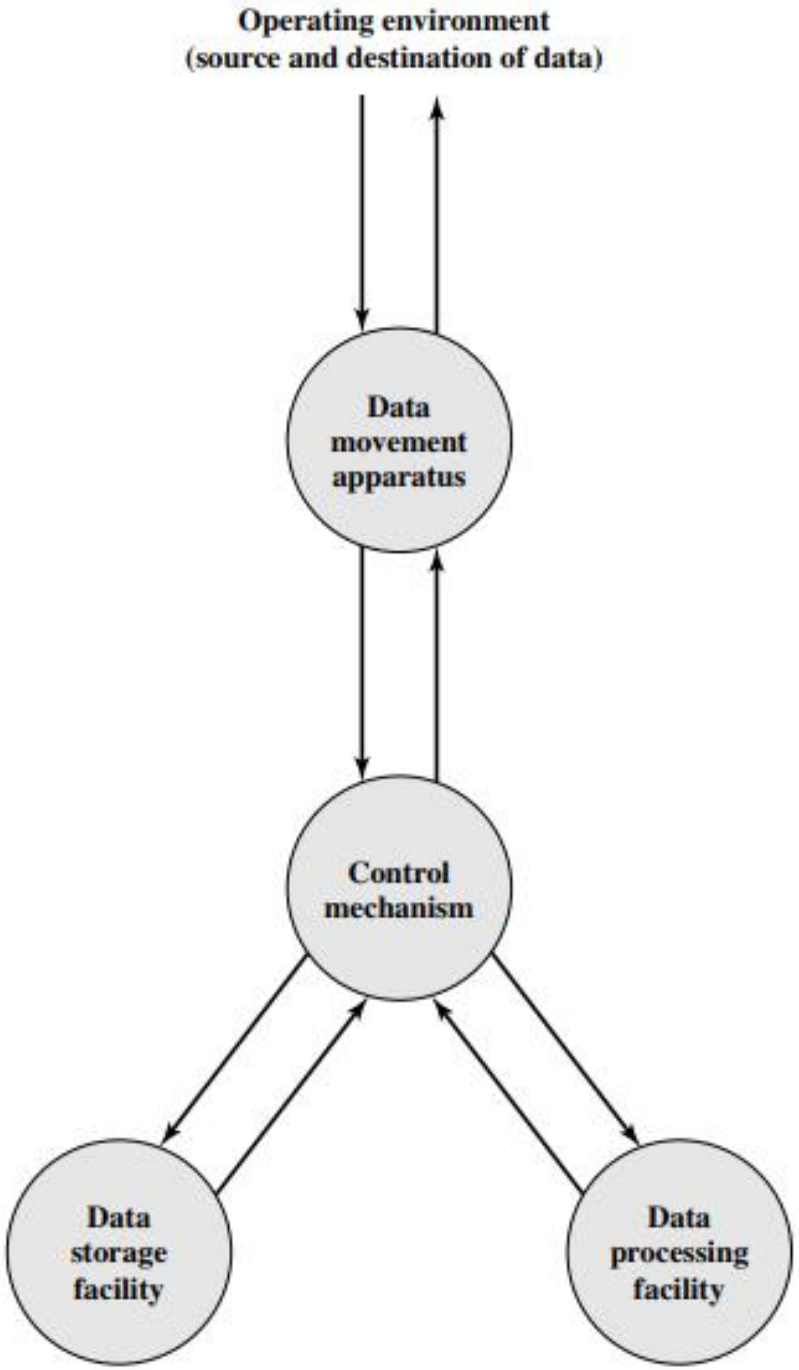
- **Structure:** The way in which the components are interrelated.
- **Function:** The operation of each individual component as part of the structure.

Function of a computer

Both the structure and functioning of a computer are simple. In general terms, there are only four basic functions that a computer can perform:

- Data processing
- Data storage
- Data movement
- Control

A Functional View of the Computer



Function

Data processing: Data may take a wide variety of forms, and the range of processing requirements is broad.

Data storage: Even if the computer is processing data on the fly (i.e., data come in and get processed, and the results go out immediately), the computer must temporarily store at least those pieces of data that are being worked on at any given moment. Thus, there is at least a short-term data storage function. Equally important, the computer performs a long-term data storage function. Files of data are stored on the computer for subsequent retrieval and update.

Function

Data movement: The computer's operating environment consists of devices that serve as either sources or destinations of data. When data are received from or delivered to a device that is directly connected to the computer, the process is known as input–output (I/O), and the device is referred to as a peripheral. When data are moved over longer distances, to or from a remote device, the process is known as data communications.

Control: Within the computer, a control unit manages the computer's resources and orchestrates the performance of its functional parts in response to instructions.

Possible Computer Operations

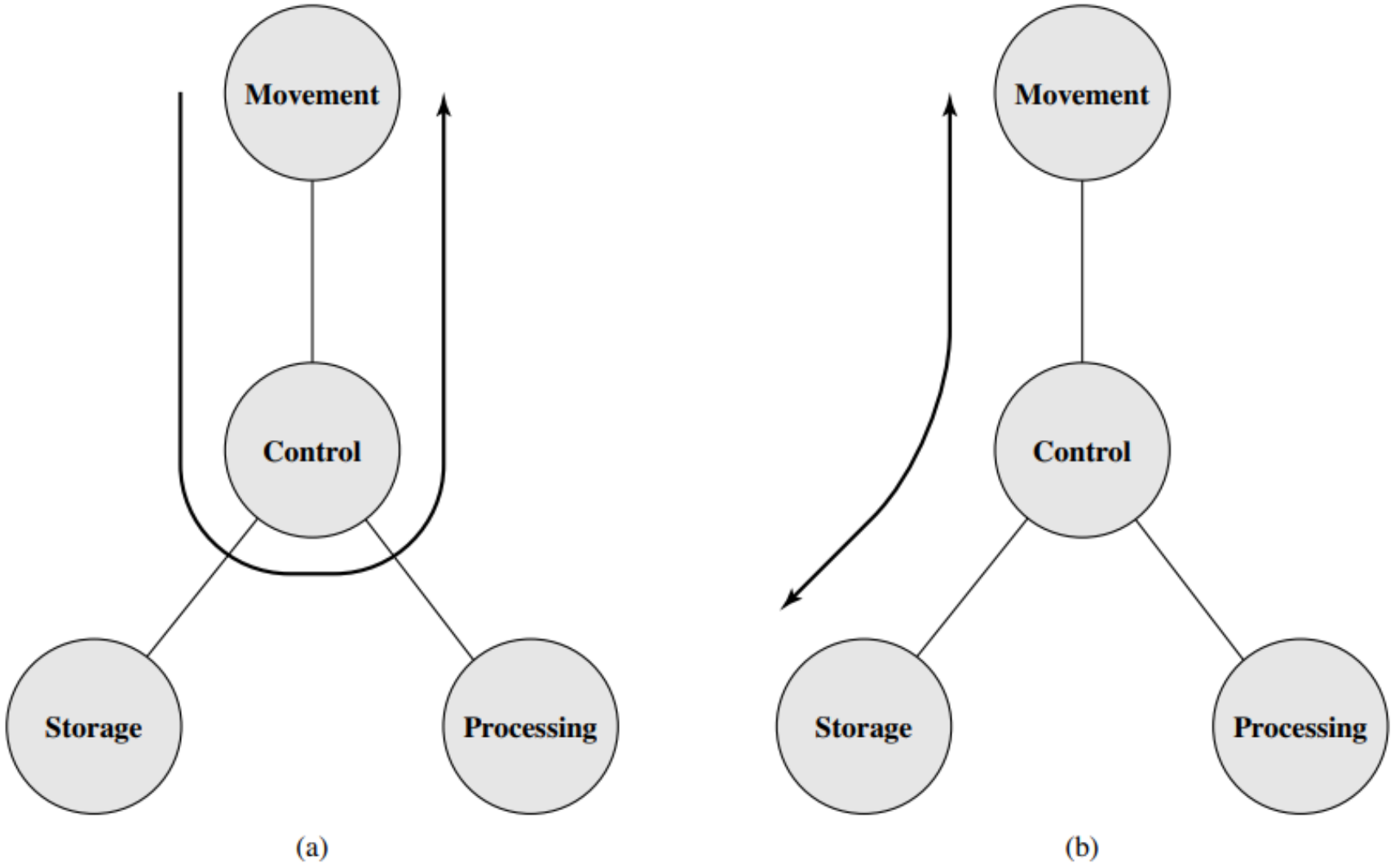
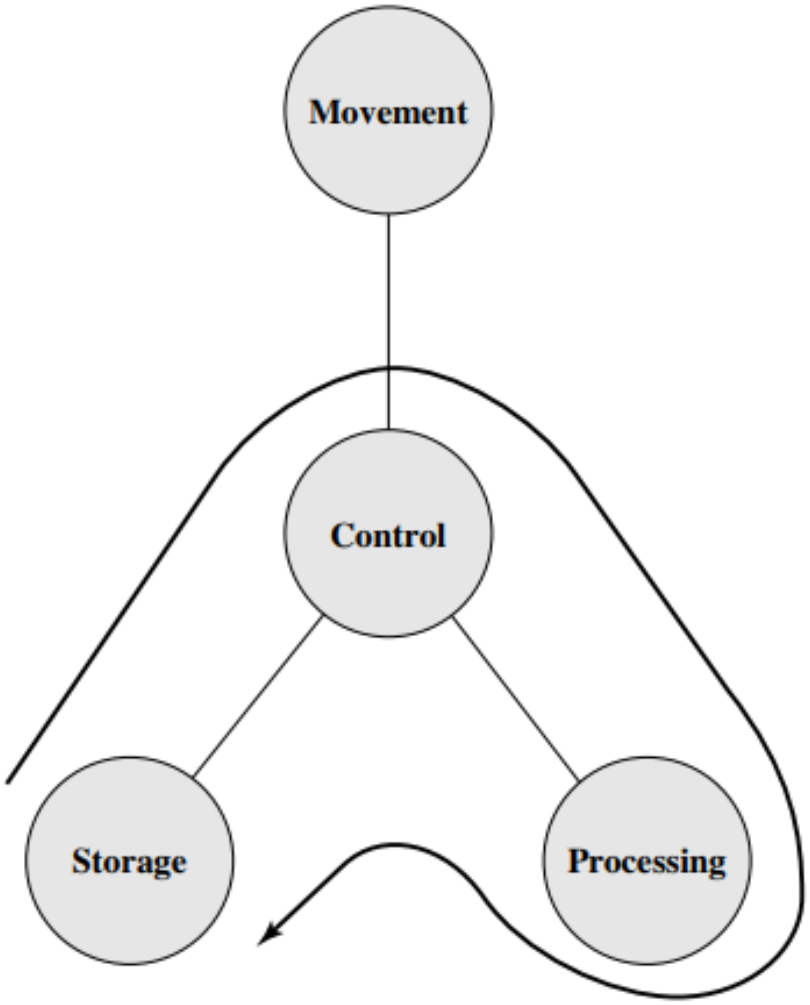


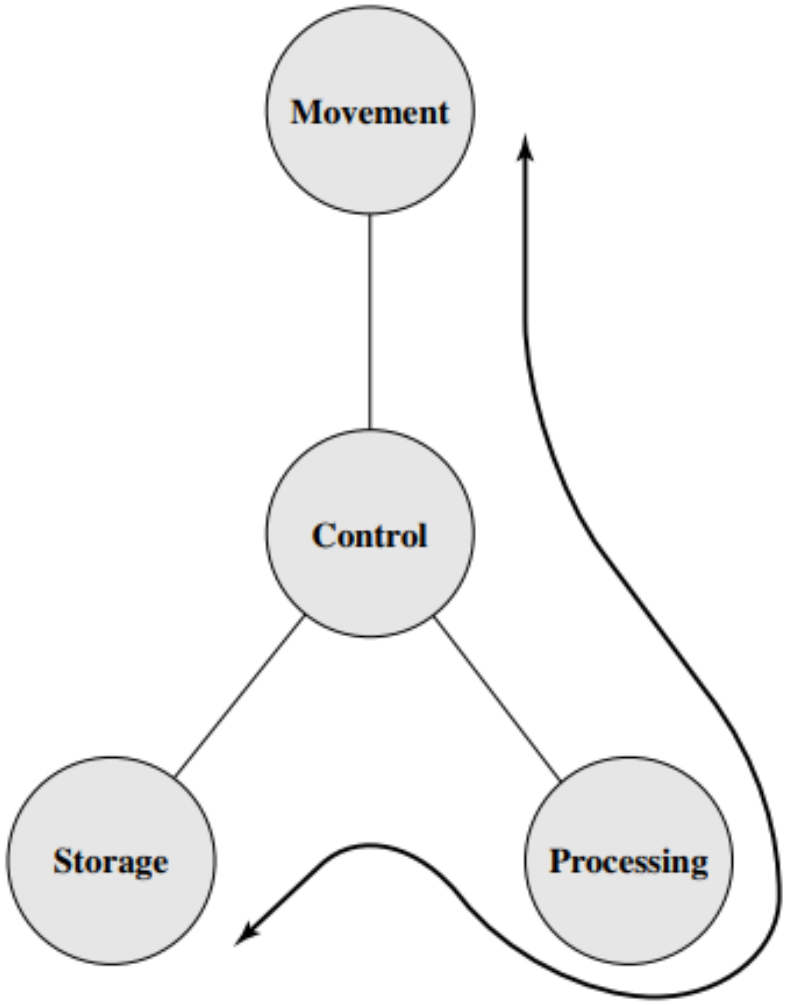
Figure a, simply transferring data from one peripheral or communications line to another. It can also function as a data storage device

Figure b, with data transferred from the external environment to computer storage (read) and vice versa (write).

Possible Computer Operations



(c)

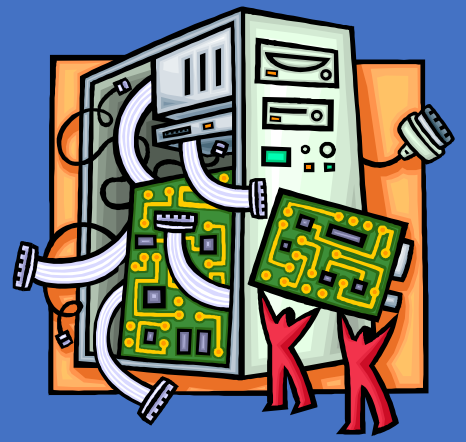


(d)

The final two diagrams show operations involving data processing, on data either in storage (Figure c) or route between storage and the external environment (Figure d).

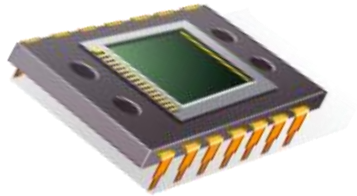
Structural View

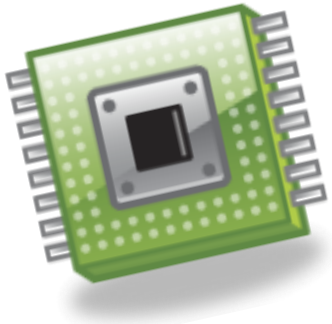
There are four main structural components of the computer:



- CPU – controls the operation of the computer and performs its data processing functions
- Main Memory – stores data
- I/O – moves data between the computer and its external environment
- System Interconnection – some mechanism that provides for communication among CPU, main memory, and I/O

Major structural components of CPU





- Control Unit
  - Controls the operation of the CPU and hence the computer
- Arithmetic and Logic Unit (ALU)
  - Performs the computer's data processing function
- Registers
  - Provide storage internal to the CPU
- CPU Interconnection
  - Some mechanism that provides for communication among the control unit, ALU, and registers

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Multicore Computer Structure

- Central processing unit (CPU)
  - Portion of the computer that fetches and executes instructions
  - Consists of an ALU, a control unit, and registers
  - Referred to as a processor in a system with a single processing unit

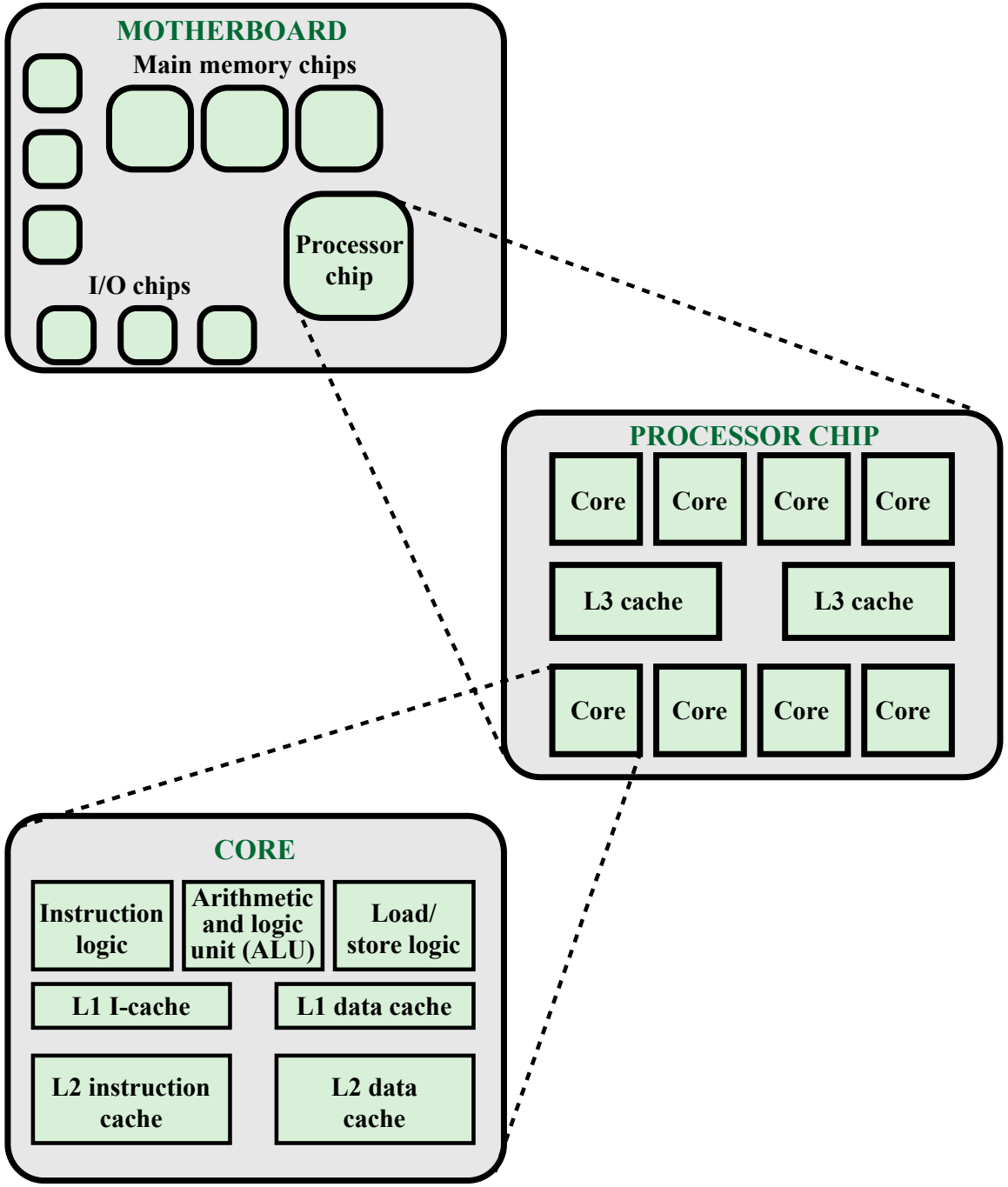
- Core
  - An individual processing unit on a processor chip
  - May be equivalent in functionality to a CPU on a single-CPU system
  - Specialized processing units are also referred to as cores

- Processor
  - A physical piece of silicon containing one or more cores
  - Is the computer component that interprets and executes instructions
  - Referred to as a *multicore processor* if it contains multiple cores

Cache Memory

- Multiple layers of memory between the processor and main memory
- Is smaller and faster than main memory
- Used to speed up memory access by placing in the cache data from main memory that is likely to be used in the near future
- A greater performance improvement may be obtained by using multiple levels of cache, with level 1 (L1) closest to the core and additional levels (L2, L3, etc.) progressively farther from the core

Figure: Simplified View of Major Elements of a Multicore Computer



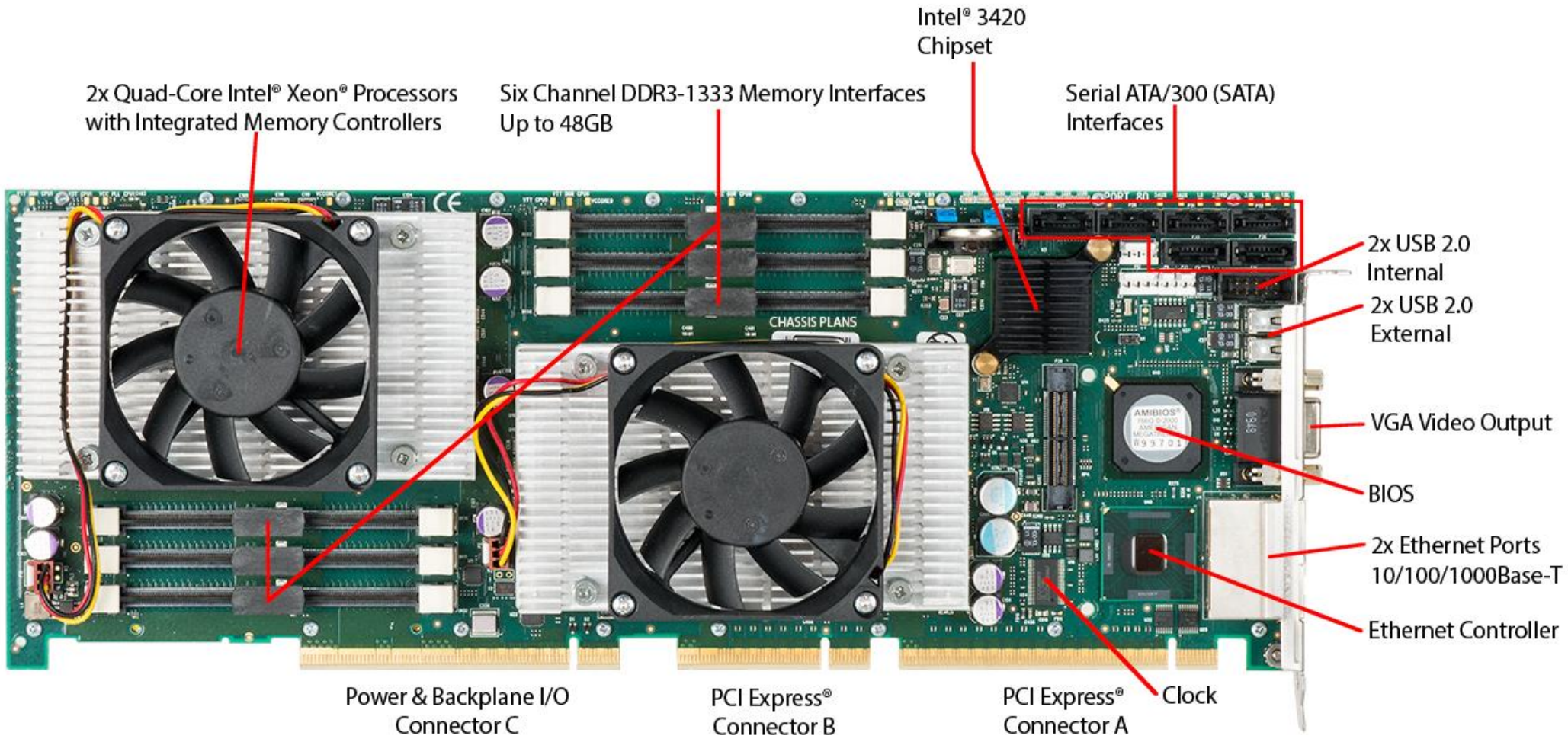
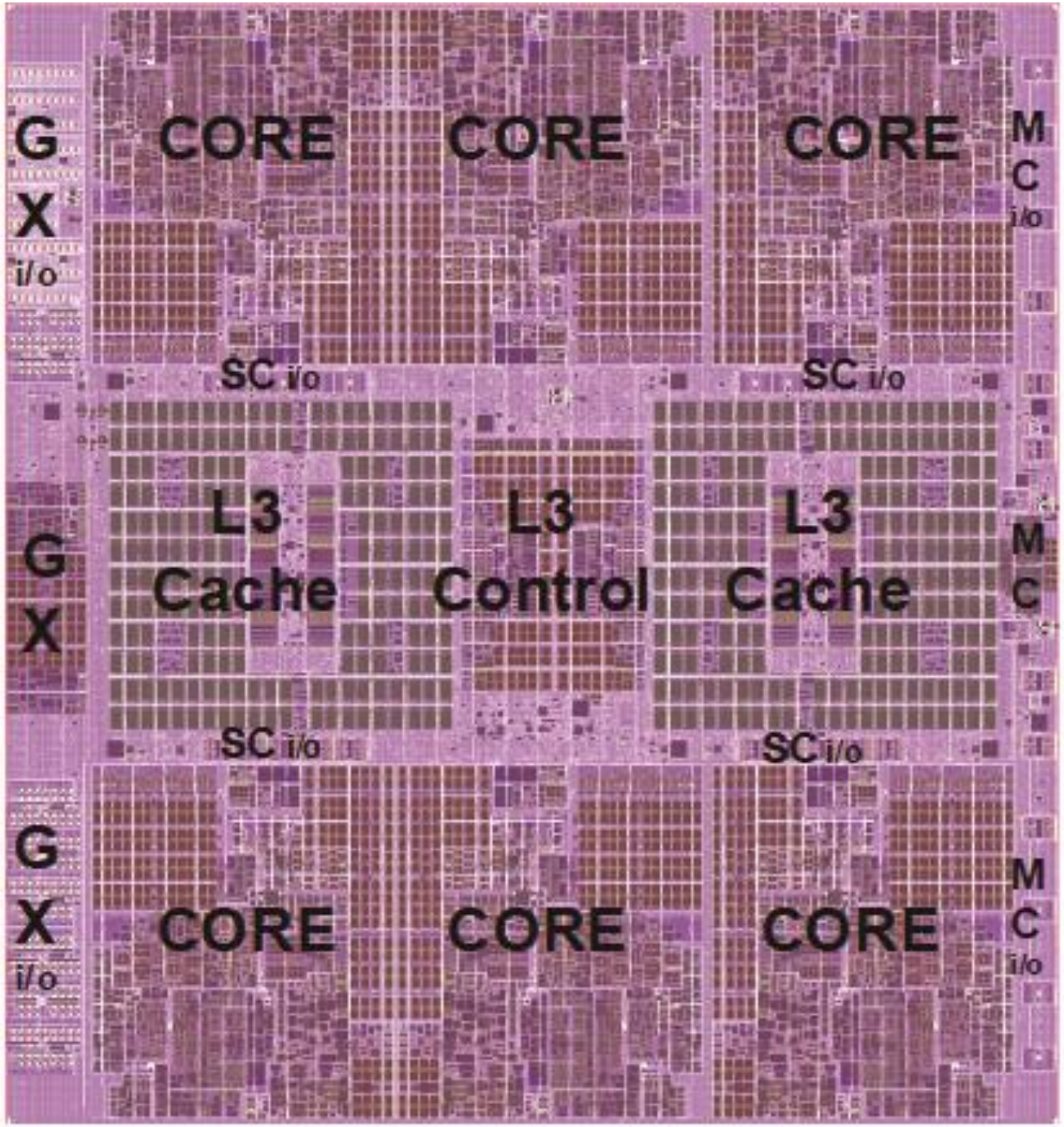


Figure: Motherboard with Two Intel Quad-Core Xeon Processors



Figure

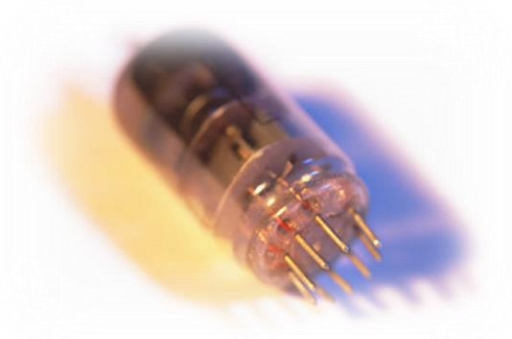
zEnterprise EC12
Processor Unit (PU)
Chip Diagram by IBM

Computer Generations

Generation	Approximate Dates	Technology	Typical Speed (operations per second)
1	1946–1957	Vacuum tube	40,000
2	1957–1964	Transistor	200,000
3	1965–1971	Small and medium scale integration	1,000,000
4	1972–1977	Large scale integration	10,000,000
5	1978–1991	Very large scale integration	100,000,000
6	1991-	Ultra large scale integration	>1,000,000,000

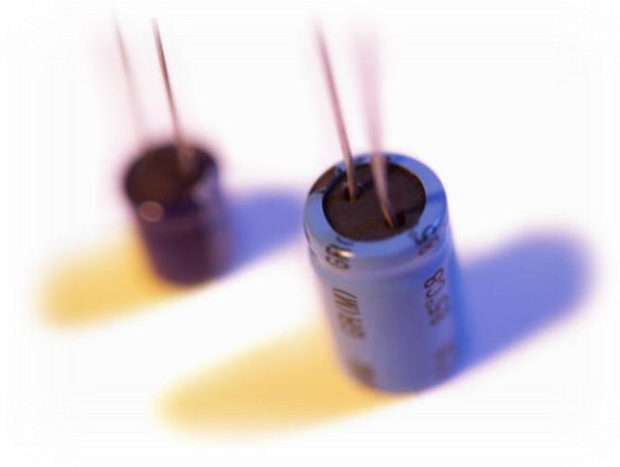
First Generation: Vacuum Tubes

- Vacuum tubes were used for digital logic elements and memory
- IAS computer
  - Fundamental design approach was the stored program concept
    - Attributed to the mathematician John von Neumann
    - First publication of the idea was in 1945 for the EDVAC
  - Design began at the Princeton Institute for Advanced Studies
  - Completed in 1952
  - Prototype of all subsequent general-purpose computers



Second Generation: Transistors

- Smaller
- Cheaper
- Dissipates less heat than a vacuum tube
- Is a *solid state device* made from silicon
- Was invented at Bell Labs in 1947
- It was not until the late 1950's that fully transistorized computers were commercially available



Third Generation: Integrated Circuits

- 1958 – the invention of the integrated circuit
- *Discrete component*
  - Single, self-contained transistor
  - Manufactured separately, packaged in their own containers, and soldered or wired together onto masonite-like circuit boards
  - Manufacturing process was expensive and cumbersome

- The two most important members of the third generation were the IBM System/360 and the DEC PDP-8

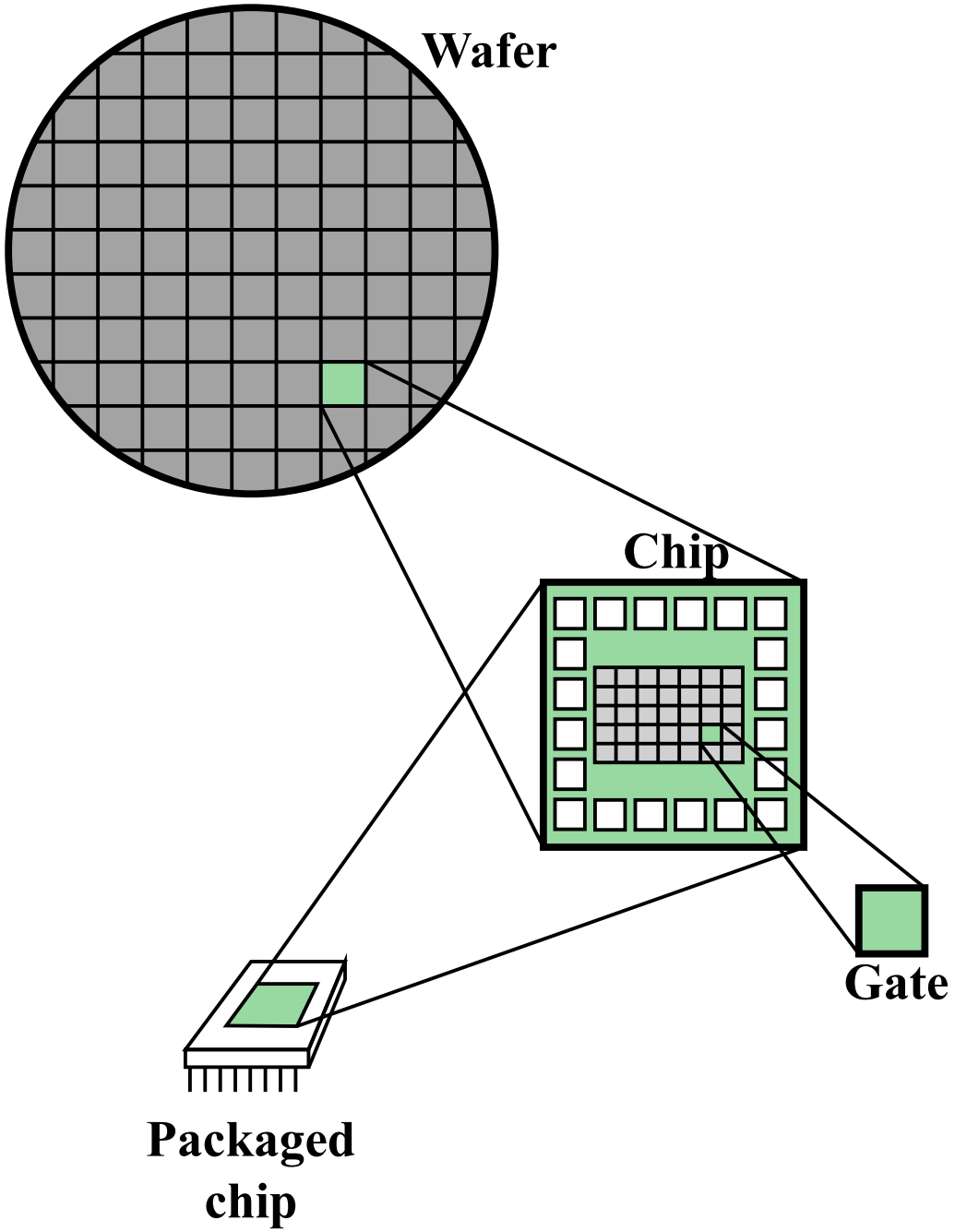
Integrated Circuits

- Data storage – provided by memory cells
- Data processing – provided by gates
- Data movement – the paths among components are used to move data from memory to memory and from memory through gates to memory
- Control – the paths among components can carry control signals

- A computer consists of gates, memory cells, and interconnections among these elements
- The gates and memory cells are constructed of simple digital electronic components

- Exploits the fact that such components as transistors, resistors, and conductors can be fabricated from a semiconductor such as silicon
- Many transistors can be produced at the same time on a single wafer of silicon
- Transistors can be connected with a processor metallization to form circuits

Relationship Among Wafer, Chip, and Gate



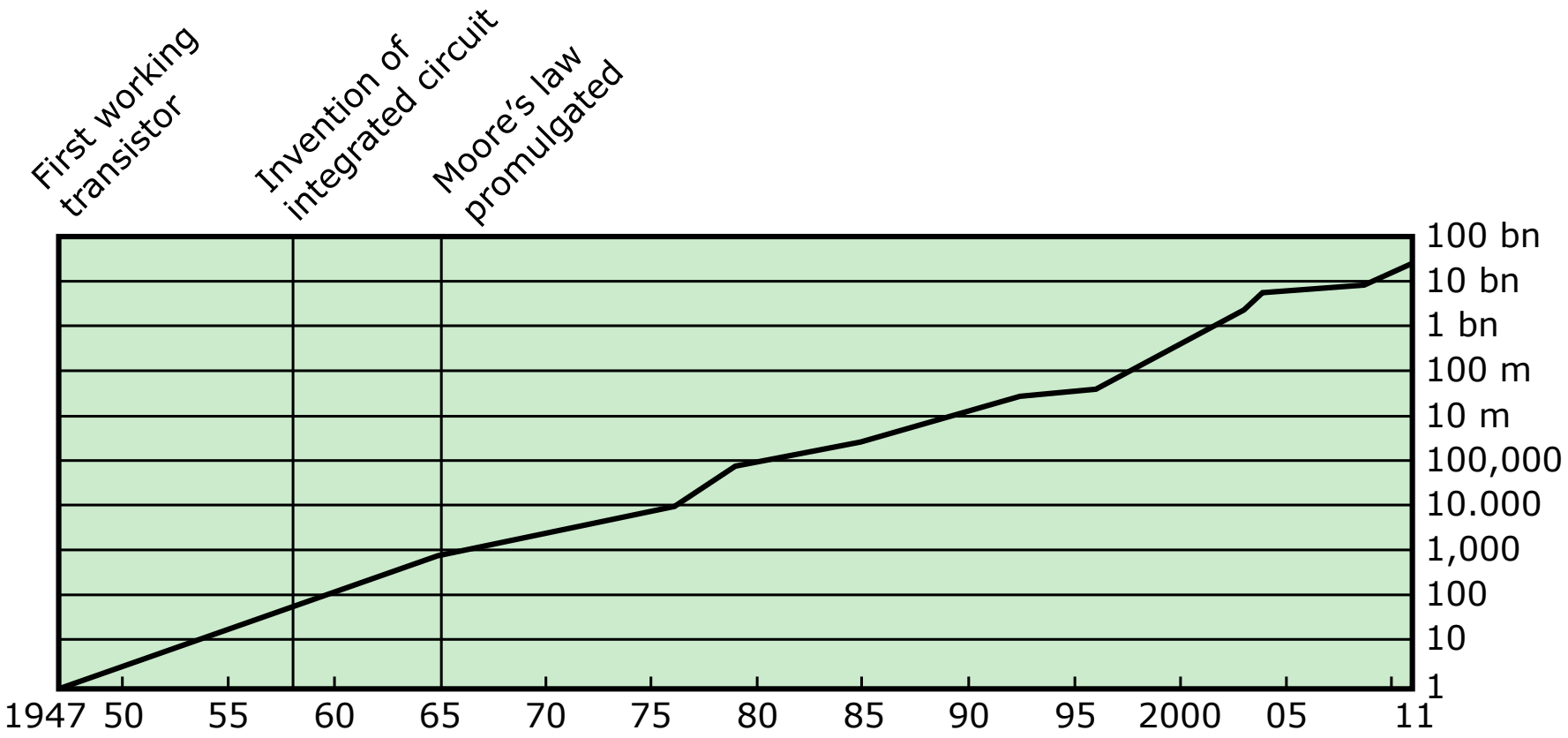


Figure 1.12 Growth in Transistor Count on Integrated Circuits (DRAM memory)